

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>First let's calculate how long it takes to download 40% of the file. The file size is 200 GB. 40% of 200 GB is $0.4 * 200 = 80$ GB. Given that Carla can download at a speed of 2 GB / minute. To download 80 GB would take $80 / 2 = 40$ minutes. After downloading 40% of the file Windows forces a restart. This restart takes 20 minutes. After the restart Carla has to restart the download from the beginning. Therefore she has to download 200 GB again. To download 200 GB at a speed of 2 GB / minute would take $200 / 2 = 100$ minutes. In total Carla needs $40 + 20 + 100 = 160$ minutes to download the entire file. </think><answer>160</answer>

